Saving and Loading Files

Making data outlive the program

THE PROBLEM

EVERYTHING SO FAR DISAPPEARS

Everything we've built lives only in the program's **memory** - it disappears the moment the program ends.

> **File I/O** is how a program remembers something between separate runs.

WRITING A FILE

`std::ofstream` ("output file stream") opens a file for writing.

```cpp
#include <fstream>

// The file is created in the working directory.
// Where the program is run from.
std::ofstream outputFile("friends.txt");

for (const std::string& friendName : friends) {
    outputFile << friendName << "\n";
}

outputFile.close();
```

Write to it with `<<`, just like `std::cout`.

READING

READING A FILE BACK

`std::ifstream` (”input file stream”) opens a file for reading.

```
std::ifstream inputFile("friends.txt");
std::vector<std::string> friendsFromDisk;

std::string line;
while (std::getline(inputFile, line)) {
    friendsFromDisk.push_back(line);
}

inputFile.close();
```

Same `std::getline` already used for keyboard input.

THE KEY IDEA

IT'S ALL JUST A STREAM

A file and the keyboard are both just a **stream of text** as far as C++ is concerned.

DIRECTION	TOOL
keyboard in	`std::cin`
console out	`std::cout` / `std::print`
file in	`std::ifstream`
file out	`std::ofstream`

DEBUG A FILE WRITE

WATCH THE FILE APPEAR

Step over the `saveFriendsToFile`-equivalent write, then open `friends.txt` in your IDE’s file browser - it now exists, with the friend names written to it.